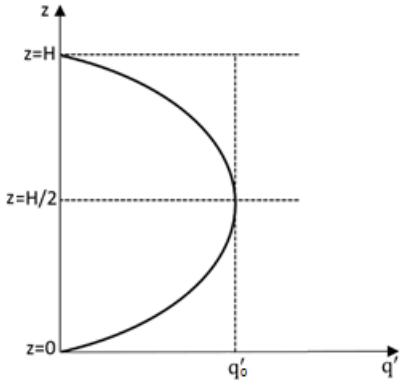
UNIVERSITY OF ILLINOIS
DEPARTMENT OF NUCLEAR, PLASMA AND RADIOLOGICAL ENGINEERING
NPRE 449 – Nuclear Systems Engineering and Design

HW8: Simplified CP

In the CP, temperatures and pressure in the hot channel of a PWR and a BWR are calculated. In both cases the linear heat generation rate changes in z as:

$$q'(z) = q'_0 \sin(\pi z/H), \quad 0 \le z \le H$$

where H is the fuel height.



Assumptions:

- Steady-state
- No internal heat generation in the fluid
- Axial conduction is negligible
- Single phase properties are constant (based on inlet pressure and temperature)
- Saturation properties (temperature, liquid saturation enthalpy, and vapor saturation enthalpy) are a function of pressure. All other properties can be treated as constant (based on inlet pressure).
- Changes in fuel and clad properties with temperature are negligible

PWR

Geometry:

- H = 4.1 m
- D_{rod} = 0.94 cm
- Pitch = 1.25 cm
- D_{Fuel} = 0.81 cm
- Gap thickness = 0.005 cm
- k_{gap}=0.25 W/m°C
- k_fuel=3.6 W/m°C;
- k_cladding=21.5 W/m°C

Conditions:

- G = 3800 kg/m^{2}s
- q'_0 = 380 W/cm
- P (z=0) = 15 MPa
- T_f (z=0) = 277°C

BWR

Geometry:

- H = 4.0 m
- D_{rod} = 1.228 cm
- Pitch = 1.621 cm
- D_{Fuel} = 1.041 cm
- Gap thickness = 0.011 cm
- k_{gap}=0.25 W/m°C
- k_fuel=3.6 W/m°C;
- k_cladding=21.5 W/m°C

Conditions:

- G = 2350 kg/m^{2}s
- q'_0 = 550 W/cm
- P (z=0) = 7.5 MPa
- T_f (z=0) = 272°C

1) For the PWR condition, assuming constant properties, derive the governing equations, calculate the parameters necessary, and provide following plots – provide copy of code with submission (NOTE: put z on the y-axis of the plot – like we have done for other reactor channel plots):
    a. Fuel Centerline temperature, clad surface temperature, liquid temperature and saturation temperature vs z
    b. Equilibrium quality vs z
    c. Pressure vs z

2) For the PWR condition, provide a sensitivity study on the changing boundary conditions and geometry on the outlet fluid equilibrium quality and max fuel centerline temperature.